

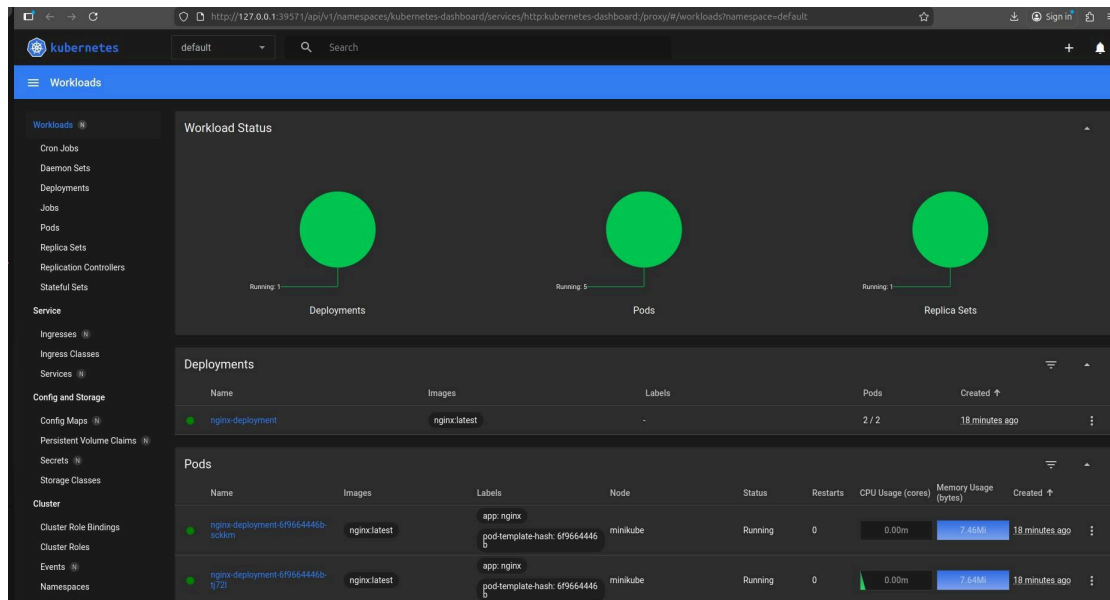
Program-10 Kubernetes Dashboard and CLI for Cluster Monitoring

Step 1: Start Minikube

```
minikube start --cpus=2 --memory=4096
```

Step 2: Enable Kubernetes Dashboard

```
minikube dashboard
```



Step 3: Create Deployment (deployment.yaml)

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/prg10
GNU nano 7.2 deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 4: Create Service (service.yaml)

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/prg10
GNU nano 7.2 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 5: Apply Resources and monitor via CLI

kubectl apply -f deployment.yaml

kubectl apply -f service.yaml

kubectl get pods

kubectl get nodes

kubectl get svc

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/prg10
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prg10$ nano deployment.yaml
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prg10$ nano service.yaml
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prg10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment unchanged
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prg10$ kubectl apply -f service.yaml
service/nginx-service created
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prg10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backup-job-29468515-4mwdq           0/1     Completed 0           12m
backup-job-29468520-bg5gn           0/1     Completed 0           7m57s
backup-job-29468525-xmr5f           0/1     Completed 0           2m57s
my-cron-29468515-mtcf9              0/1     Completed 0           12m
my-cron-29468520-78dvt              0/1     Completed 0           7m57s
my-cron-29468525-w7xzs              0/1     Completed 0           2m57s
nginx-deployment-6f9664446b-n6d2r   1/1     Running   2 (3m41s ago)  39m
nginx-deployment-6f9664446b-z2vs6   1/1     Running   2 (3m41s ago)  39m
nodejs-pod                          1/1     Running   8 (3m31s ago)  31d
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prg10$ kubectl get svc
NAME            TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes      ClusterIP   10.96.0.1    <none>        443/TCP          23d
nginx-deployment NodePort    10.104.20.56 <none>        80:31678/TCP     38m
nginx-service   NodePort    10.102.213.85 <none>        80:31046/TCP     31s
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prg10$ kubectl get nodes
NAME        STATUS   ROLES    AGE   VERSION
minikube    Ready   control-plane 48d   v1.34.0
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/prg10$
```

Step 6: Access Application

minikube service nginx-service

Opens NGINX welcome page in browser.

